

MISSION: EIFFEL TOWER ECOTRAIL PARIS 120KM

Tactical Roadbook & Crew Manual

- **ATHLETE:** -
- **CREW:** -
- **TARGET:** -
- **DATE:** 21 March 2026 -

"The race begins at KM 80. Everything before that is just the commute"

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Introduction

The Philosophy: The Day-to-Night Mission

The strategy is built on a single competitive advantage: **Conserving Road Speed**. Your marathon personal best allows you to maintain an efficient, low-stress aerobic pace that is significantly faster than the average trail runner's "easy" effort.

The core objective of the first 40km is to "bank" a safety margin of approximately 45–60 minutes. This is not for ego; it is a tactical buffer designed to absorb the inevitable technical slowdowns of the Chevreuse valleys and the navigation overhead of the night forests of Meudon and Saint-Cloud.

- **Phase A (KM 0–40):** Flow and Cadence. Use the Rambouillet allées to establish a high turnover (175+ spm) while maintaining a low heart rate (RPE 4/10).
- **Phase B (KM 40–80):** Resistance. Transition to technical climbing. The goal shifts from "gaining time" to "protecting the engine." Walk the steep ramps, run the crowns.
- **Phase C (KM 80–120):** The Dark Shift. Focus on movement economy, lighting management, and pushing through systemic fatigue.

Emergency Protocol

In the heat of the race, logic fades. These protocols and contacts are absolute and must be followed by the crew if the runner is unable to make clear decisions.

CRITICAL CONTACTS & INFO

- **Race HQ Emergency:**
- **Lead Crew:**
- **Runner ICE:**
- **Medical Disclosure:** No known allergies.
- [Website](#) and [rules](#).

Safety & Resilience: Validation for the Crew

Special Note for the Crew:

It is a logical fallacy to view 120km as a purely destructive act. Modern sports science and specific preparation present a process of biological hardening.

1. **Mechanotransduction & Joint Health:** Rhythmic, long-distance impact actually *thickens* knee cartilage and increases bone mineral density in trained athletes. The runner is not "wearing out" his joints; he is making them more resilient.
2. **Cardiac Efficiency:** A low resting heart rate (RHR) is a marker of an elite cardiovascular system. This heart is structurally prepared to handle high-volume endurance without pathological stress.
3. **The Risk Factor:** The primary safety risk is not injury, but **Metabolic Exhaustion** (sugar/salt levels). This is where your role is critical. If he is fueled and hydrated, he is safe.
4. **The Mission:** The runner is realizing the potential of a body he has spent years training. Your presence is his safety net and his "external brain" for the final 40km.

The External Brain: Crew Manual

The Mission: Externalizing Cognitive Load

In an ultra-endurance event, the athlete eventually loses the ability to make logical decisions due to glycogen depletion and sleep deprivation. **You are the External Brain.**

Your role is to give clear, non-negotiable instructions. Do not ask "How are you feeling?" or "What do you want to eat?". Instead, state: "Drink this now," "Eat two crackers," or "You are leaving in 60 seconds."

Communication: The "Silent Mile" Protocol

To ensure the aid station transition is seamless, we use the following signaling:

- **The 1km Signal:** I will send a single-word text (e.g., "SOUP", "FEET", "COLD").
- **Your Response:** Text me your exact location in the crew zone (e.g., "I am 10m past the entrance, right side, wearing the blue hat").
- **Visual Audit:** Upon my arrival, look at my eyes. If they are glazed, prioritize immediate liquid sugar (Coke) before any other gear tasks.

The 5-Minute Pitstop Protocol

Efficiency at aid stations is how we maintain the 13h pace. Every station should follow this timed sequence:

1. **Minute 1: The Dump & Fill.** Take my trash. Hand me the fresh, uncapped soft flasks.
2. **Minute 2: The Tactical Review.** Hold up the A4 "Deep Dive" sheet for the *next* segment. Point out the pace target and key hazards.
3. **Minute 3: Mechanical Check.** Apply anti-chafe or spray magnesium if requested. Change batteries if it is a "Battery Swap" station.
4. **Minute 4: Forced Calorie Loading.** Ensure I consume at least 200 calories before standing up.
5. **Minute 5: The Ejection.** Physically help me stand up and point me toward the exit.

Troubleshooting & DNF Logic

CREW DECISION TREE

- **If Nausea:** Stop gels. Give plain water. Wait 20 mins. Reintroduce salt.
- **If Mental Collapse:** Remind me: "The race starts at KM 80. You are just commuting right now."
- **DNF Authority:** We only stop for **Structural Failure** (bone pain, inability to bear weight) or **Medical Emergency** (total disorientation). Muscle pain is standard; we keep moving.

Crew Essential Loadout

Ensure you have the following in the car for your own performance:

- **Power:** Phone power bank and car charger.
- **Comfort:** A warm down jacket (waiting is cold) and comfortable waterproof shoes.
- **Docs:** This Master Roadbook (Physical copy) and the Pace Chart on the dashboard.

NLDR: I recommend fine-tuning the Gear appendix to better prepare the runner's and crew's loadout.

Route

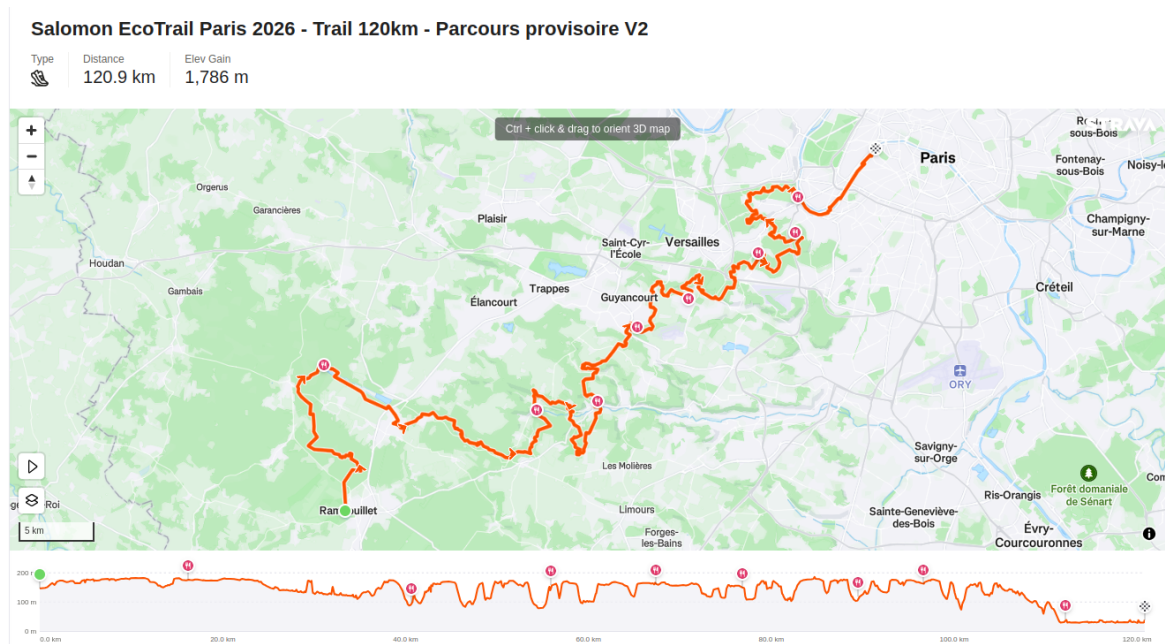


Figure 1: EcoTrail Paris 120km Course Map.

Notes

EcoTrail Paris 2026 - 120km Roadbook Reference

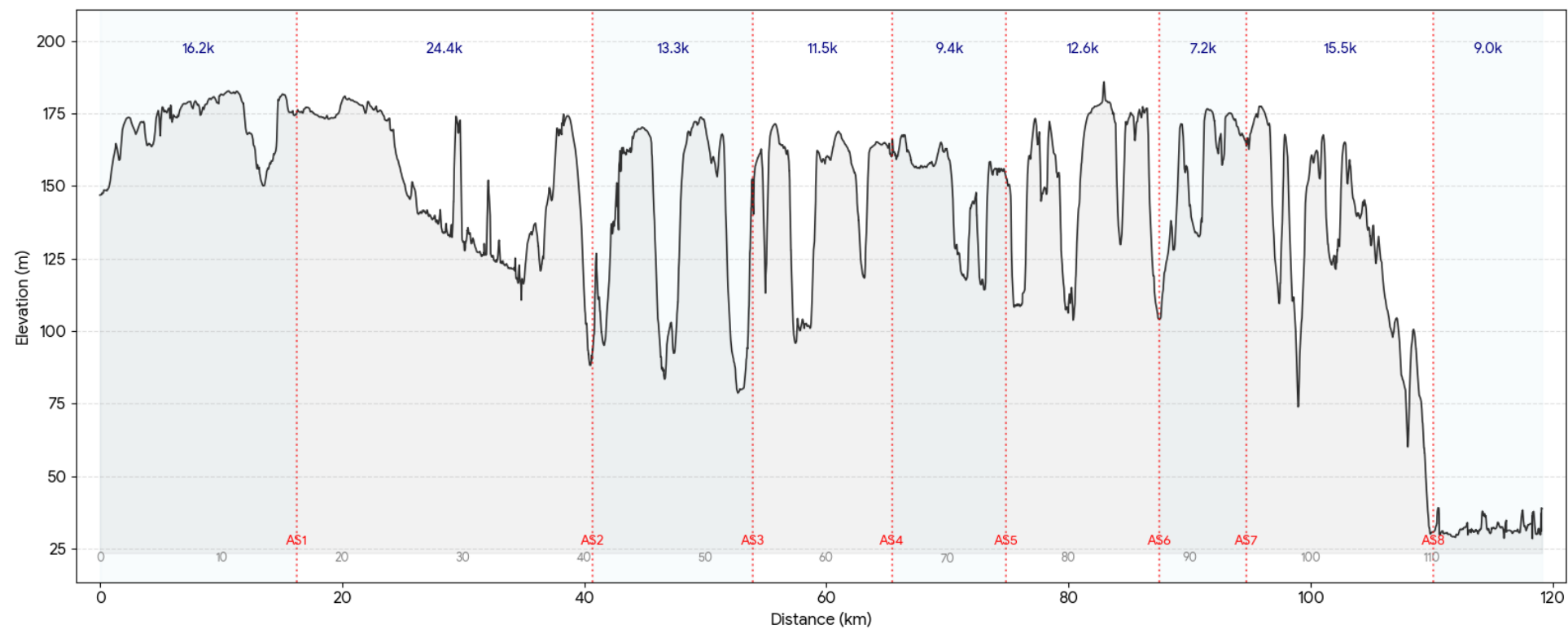
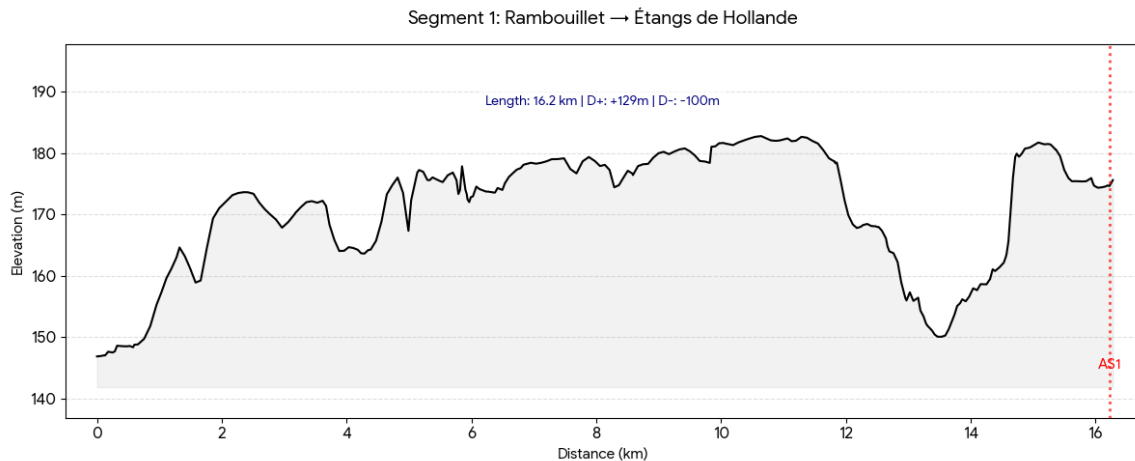


Figure 2: Full Course Elevation Profile - EcoTrail Paris 2026

Segment 1: Rambouillet → Étangs de Hollande



Dist: 0 → 16.2 km

Pace: 5:15–5:45

Time: Start → +1h30

Terrain: Forest Track

Elev: +140m/-120m

RPE: 4/10

Terrain Analysis

- **Surface Composition:** 85% hard-packed forest fire roads, 10% tarmac, 5% loose sand.
- **Profile:** Net neutral. Initial 5km climb to a plateau, followed by rolling undulations and a final descent to the ponds.
- **Technicality:** Low. High-speed section. Risk of congestion in the first 2km leaving the start area.

Critical Hazards

- **The Sand Pit:** Approximately KM 8-10. Patches of soft sand can cause sudden heart rate spikes if pace is forced.
- **The Speed Trap:** The firm surface invites a road-running gait. Excessive impact here leads to early quad fatigue and tibial stress later in the race.
- **Early Dehydration:** Because it is early and likely cool, runners often forget to start the 500ml/hour protocol.

Nutrition and Hydration

Nutrition: Target 40-60g Carbs/hour. 500-700ml fluids with electrolytes based on temperature.

Aid Station Actions:

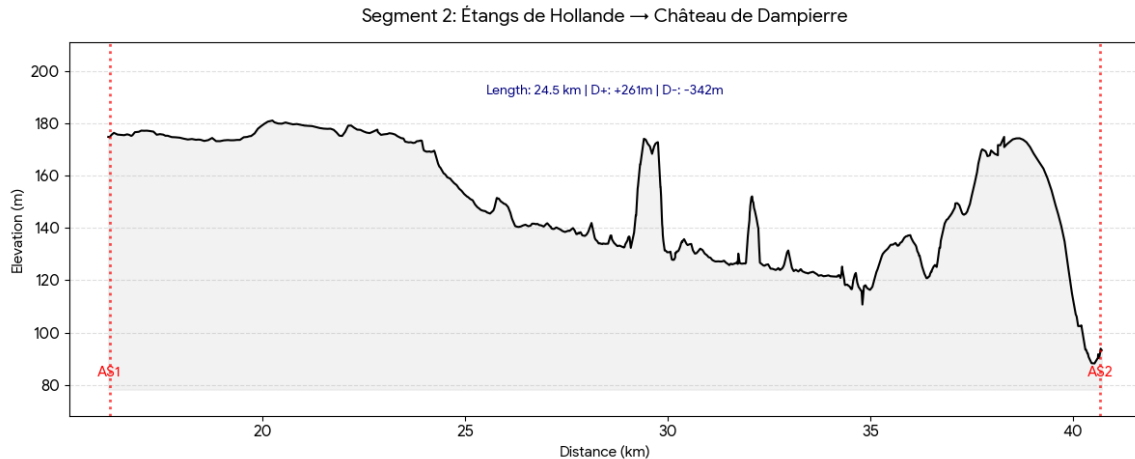
Refill: Compulsory 1L refill (even if partially full) to prepare for the long S2 stretch.

Salt: First intake of sodium if sweat rate is high.

Efficiency: AS1 is often crowded. Aim for a <3 minute transition.

Up Next: S2: Étangs → Dampierre (24.4km). *Warning: Longest segment of the race with significant elevation gain.*

Segment 2: Étangs de Hollande → Château de Dampierre



Dist: 16.2 → 40.7 km

Pace: 6:00–6:45

Time: +1h30 → +3h50

Terrain: Technical Trail

Elev: +480m/-510m

RPE: 7/10

Terrain Analysis

- **Surface Composition:** 60% single-track forest paths, 30% dirt roads, 10% technical descents with roots/rocks.
- **Profile:** High undulation. This segment enters the **Vallée de Chevreuse**. Expect "sawtooth" climbing patterns.
- **Technicality:** Moderate-High. If wet, this section becomes high-hazard for slips.

Critical Hazards

- **The Root Garden:** Descending into Dampierre involves exposed roots. High ankle-roll risk.
- **Eccentric Loading:** The cumulative vertical gain begins to fatigue the quadriceps here. Excessive braking on descents will jeopardize the final 40km of the race.

Nutrition and Hydration

Nutrition: Target 60g Carbs/hour. High electrolyte concentration to prevent cramping during heavy climbing.

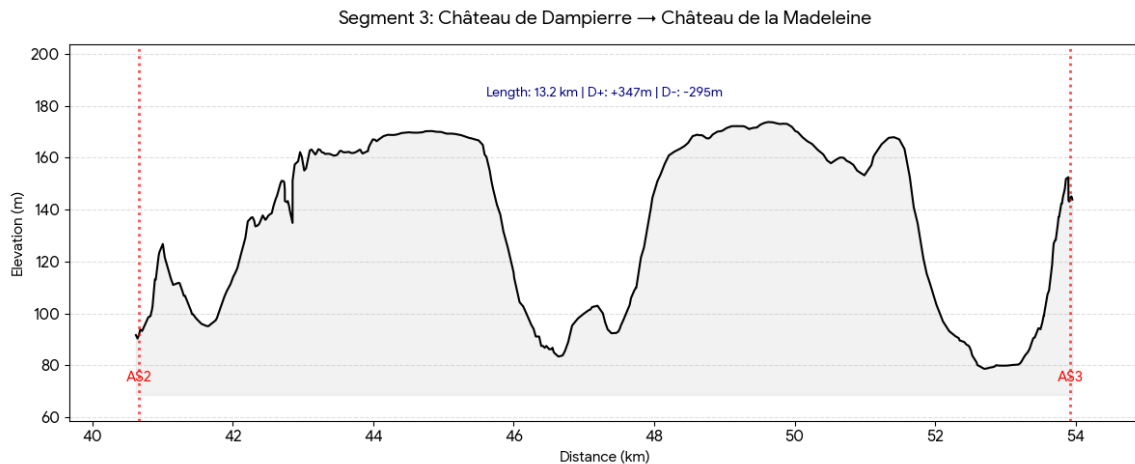
Aid Station Actions:

Full Resupply: S3 is a shorter gap, but S2 is the highest energy expenditure per km.

Solid Check: Assess stomach comfort before the steep climbs of Segment 3.

Up Next: S3: Dampierre → Château de la Madeleine (13.3km). Short but steep.

Segment 3: Château de Dampierre → Château de la Madeleine



Dist: 40.7 → 53.9 km
Pace: 6:30–7:15

Time: +3h50 → +5h30
Terrain: Technical/Steep Trail

Elev: +348m/-296m
RPE: 7/10

Terrain Analysis

- **Surface Composition:** 70% technical single-track, 20% forest floor (leaf litter/mud), 10% cobblestones/stairs near the Château.
- **Profile:** Aggressive "Sawtooth." This segment features some of the steepest localized gradients of the race. The final approach to AS3 involves a sustained technical climb.
- **Technicality:** High. Significant root density and uneven steps.

Critical Hazards

- **The Quadriceps Burn:** The repetitive nature of short, steep climbs followed by immediate descents prevents a steady rhythm and increases the risk of early-onset cramping.
- **Technical Fatigue:** By KM 50, concentration may begin to waver. The descents in this section require precise foot placement to avoid "toe-stub" trips on hidden rocks or roots.
- **Temperature Traps:** If sunny, the valley floor can be significantly more humid than the plateaus.

Nutrition and Hydration

Nutrition: Target 60g Carbs/hour. High electrolyte is vital to manage the high vertical workload.

Aid Station Actions:

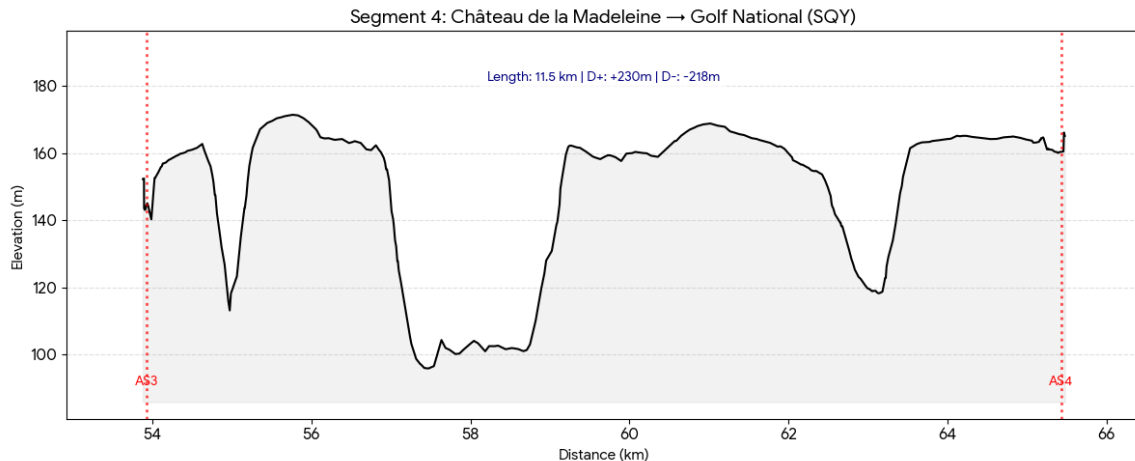
Salt Intake: Essential sodium check-in. The technical climbing will increase sweat rates even at lower air temperatures.

Solid Food: Prioritize easily digestible solids (potatoes/banana) at AS3 to prepare for the transition to more runnable ground in S4.

Mental Reset: Use the view at Château de la Madeleine as a mental landmark for completing the first technical "third" of the race.

Up Next: S4: Château de la Madeleine → Golf National (11.5km). Faster, flatter ground ahead.

Segment 4: Château de la Madeleine → Golf National (SQY)



Dist: 53.9 → 65.4 km

Pace: 5:30–6:15

Time: +5h30 → +6h45

Terrain: Mixed / Open Tracks

Elev: +231m/-218m

RPE: 5/10

Terrain Analysis

- **Surface Composition:** 50% wide dirt paths, 30% cycle lanes/tarmac, 20% technical forest exits.
- **Profile:** Gradually flattening. The initial descent from the Château is technical, followed by long, straight sections with very low technical difficulty.
- **Technicality:** Low. This is the fastest segment since the start.

Critical Hazards

- **Wind Exposure:** The area around the Golf National and the SQY plateau is often highly exposed. If running into a headwind, tuck into a group or drop the pace to avoid burning matches.
- **The "Roadie" Temptation:** After 50km of trails, the asphalt sections feel "free." Do not exceed your aerobic threshold. Maintain 2h38 marathon discipline—efficiency over ego.
- **Monotony:** The long straights around the Golf can be mentally draining compared to the engaging technicality of the previous segments.

Nutrition and Hydration

Nutrition: Target 50g Carbs/hour. Easier swallowing of solids compared to steep climbs.

Aid Station Actions:

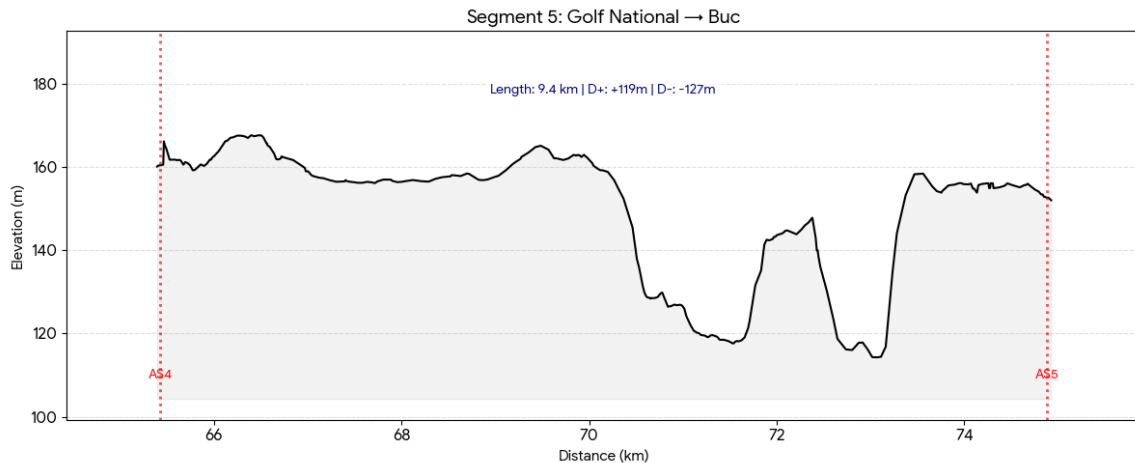
The Halfway Check: You are roughly at KM 65. Evaluate digestive comfort. Switch to liquid calories (Isotonic) if solids feel "heavy" in the stomach.

Hydration Audit: Check total fluid consumed vs. distance. If behind, use this flatter section to catch up on water intake.

Footcare: AS4 is a good point to check for hot spots or grit in the shoes before entering the urban-forest transition.

Up Next: S5: Golf National → Buc (9.4km). Transition into the final third of the race.

Segment 5: Golf National (SQY) → Buc



Dist: 65.4 → 74.9 km

Pace: 5:45–6:30

Time: +6h45 → +7h50

Terrain: Urban-Forest Mix

Elev: +119m/-127m

RPE: 5/10

Terrain Analysis

- **Surface Composition:** 60% suburban gravel tracks and paved park paths, 40% enclosed forest single-track.
- **Profile:** Moderate undulation. The segment features a gradual climb out of the SQY basin followed by a rolling descent into Buc.
- **Technicality:** Low-Moderate. Trails are generally wider but can become twisting as they approach the urban interface.

Critical Hazards

- **Navigation Micro-Turns:** This segment often involves frequent direction changes. Fatigue at KM 70 can lead to missed trail markers; stay vigilant.
- **Transition Fatigue:** The change from long open straights (S4) to more technical, enclosed woods can disrupt your pacing rhythm.
- **Light Transition:** Depending on your start wave and pace, this may be the segment where natural light begins to fade. Ensure your headlamp is reachable.

Nutrition and Hydration

Nutrition: Target 50g Carbs/hour. Maintain steady electrolyte intake to prevent late-race muscle spasms.

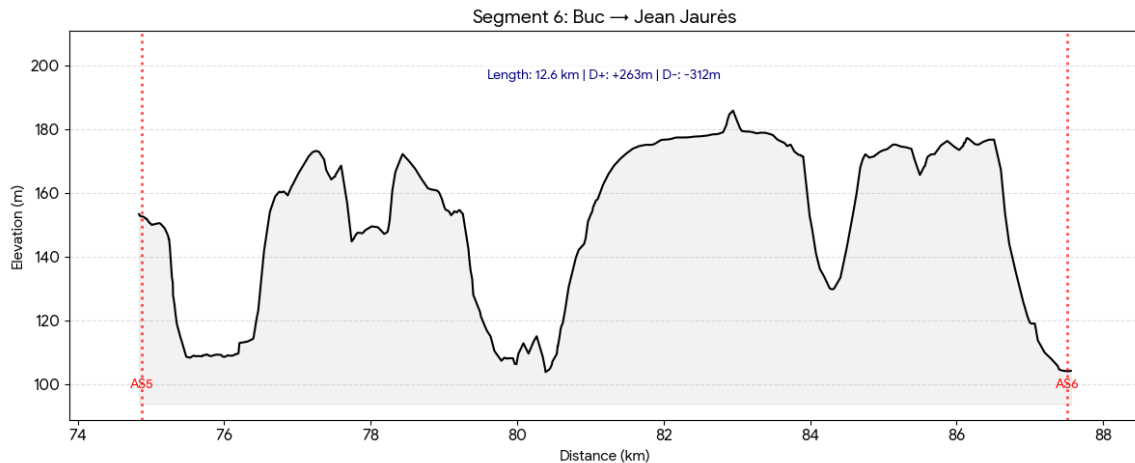
Aid Station Actions:

Energy Audit: AS5 (Buc) is the gateway to the final 45km. Check your "energy bank." If feeling hollow, prioritize complex carbohydrates at the station.

Hydration Check: Ensure you are not over-drinking pure water without electrolytes (Hyponatremia risk).

Up Next: S6: Buc → Jean Jaurès (12.6km). Entering the rolling hills of the inner forest loops.

Segment 6: Buc → Jean Jaurès



Dist: 74.9 → 87.5 km

Pace: 6:15–7:00

Time: +7h50 → +9h15

Terrain: Rolling Forest Loops

Elev: +264m/-313m

RPE: 6/10

Terrain Analysis

- **Surface Composition:** 80% forest paths, 15% compact gravel, 5% urban sidewalk.
- **Profile:** Constant rolling hills. While no individual climb is extremely long, the cumulative frequency of these "bumps" can be mentally taxing after 80km.
- **Technicality:** Moderate. Well-maintained paths, but can become extremely slippery and "clogged" with mud in wet conditions.

Critical Hazards

- **The "Ghost" Hills:** Small, punchy climbs that aren't significant on a map but feel triple their size due to accumulated fatigue.
- **Night Navigation:** For most runners, this segment will be tackled in full darkness. Sharp turns and micro-undulations require a high-lumen headlamp to judge depth and avoid tripping.
- **Leg Stiffness:** The net descent (-313m) consists of many small drops. Avoid "slamming" your feet; maintain a soft, mid-foot strike to protect your joints for the final 30km.

Nutrition and Hydration

Nutrition: Target 50–60g Carbs/hour. Prioritize liquid nutrition if nausea begins to set in.

Aid Station Actions:

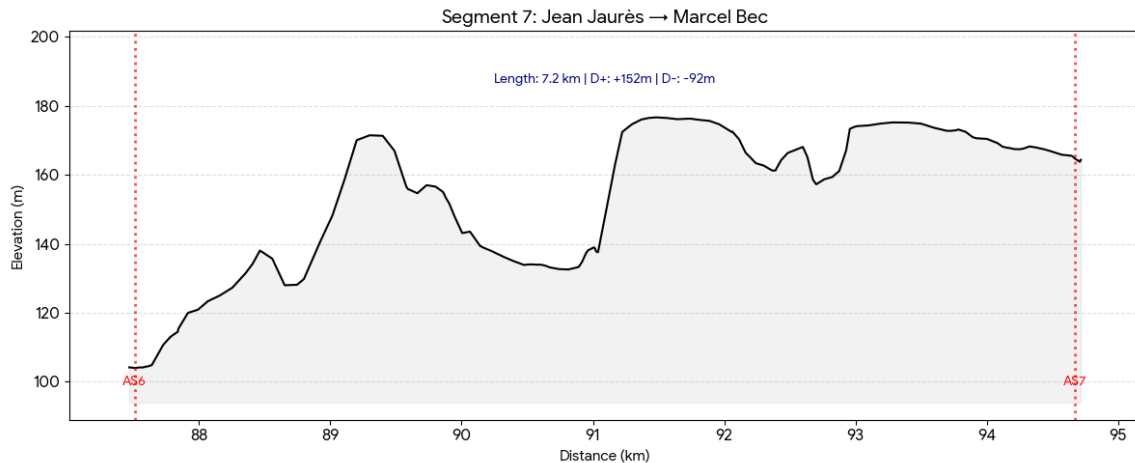
The "80km Wall": This is a common point for digestive distress. Sip small amounts frequently rather than gulping.

Caffeine Strategy: If planned, this is often a prime window for a caffeine boost to combat nighttime mental fog.

Full Resupply: AS6 is critical. Ensure a complete liquid and electrolyte restock.

Up Next: S7: Jean Jaurès → Marcel Bec (7.2km). Short but technically demanding section.

Segment 7: Jean Jaurès → Marcel Bec



Dist: 87.5 → 94.7 km

Pace: 7:00–8:00

Time: +9h15 → +10h15

Terrain: Technical Forest Trails

Elev: +153m/-93m

RPE: 8/10

Terrain Analysis

- **Surface Composition:** 75% forest single-track, 20% technical descents (roots/mud), 5% gravel.
- **Profile:** Sharp and technical. The segment begins with a significant sustained climb out of the Jean Jaurès valley.
- **Technicality:** High. This section contains some of the most "authentic" trail running of the course, often compounded by darkness and fatigue.

Critical Hazards

- **Footing Failure:** At KM 90, the proprietary receptive system is fatigued. The uneven forest floor and exposed roots in this segment present the highest risk of ankle and knee injuries.
- **The "Mental Dip":** Because the distance between these aid stations is short (7.2km), runners often underestimate the time required. Expect slow progress.
- **Restricted Passing:** Narrow single-track sections may lead to frustration if stuck behind slower runners. Use wide sections or fire-breaks to pass safely.

Nutrition and Hydration

Nutrition: Maintain 60g Carbs/hour. Liquid carbohydrates are highly recommended if solid food is no longer tolerated.

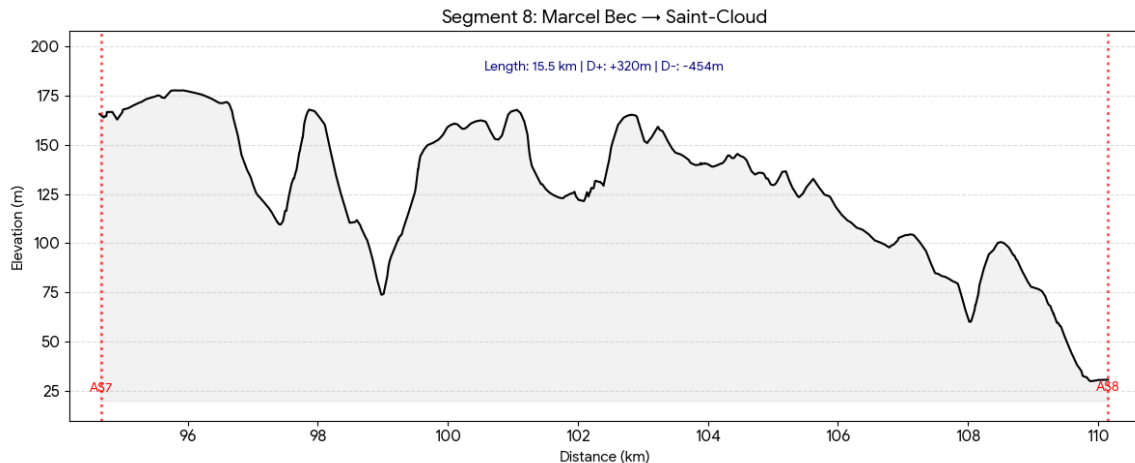
Aid Station Actions:

Concentration Check: Use small sips of strong electrolyte mix to maintain cognitive function through the technical woods.

Temperature Management: Marcel Bec is often an exposed, high-wind aid station. Prepare to layer up before leaving if the night is cold.

Up Next: S8: Marcel Bec → Saint-Cloud (15.5km). The final major forest traverse before the descent into Paris.

Segment 8: Marcel Bec → Saint-Cloud



Dist: 94.7 → 110.2 km

Pace: 6:45–7:45

Time: +10h15 → +12h15

Terrain: Forest & Park Descent

Elev: +320m/-455m

RPE: 8/10

Terrain Analysis

- **Surface Composition:** 70% forest fire roads and wide paths, 20% paved park alleys (Saint-Cloud), 10% technical forest edge.
- **Profile:** Net Descent. While the overall trend is downward (-455m), it is interrupted by several sharp, late-race climbs, particularly as you enter and traverse the heights of Saint-Cloud.
- **Technicality:** Moderate. The paths are generally wider than S7, but the descent into Saint-Cloud can be fast and hard on fatigued joints.

Critical Hazards

- **The "False" End:** The descent into Saint-Cloud often tricks runners into thinking the work is done. Significant, steep climbs remain within the park that can break a runner mentally if not expected.
- **Joint Impact:** The significant cumulative descent (-455m) at KM 100+ is a primary cause of knee and hip pain. Maintain core engagement to absorb shock.
- **Peripheral Vision:** In late-night conditions, tunnel vision is common. Force yourself to scan for markers, as Saint-Cloud has many intersecting paths.

Nutrition and Hydration

Nutrition: Target 40–50g Carbs/hour. High electrolyte concentration is essential to prepare for the final flat 9km "time trial."

Aid Station Actions:

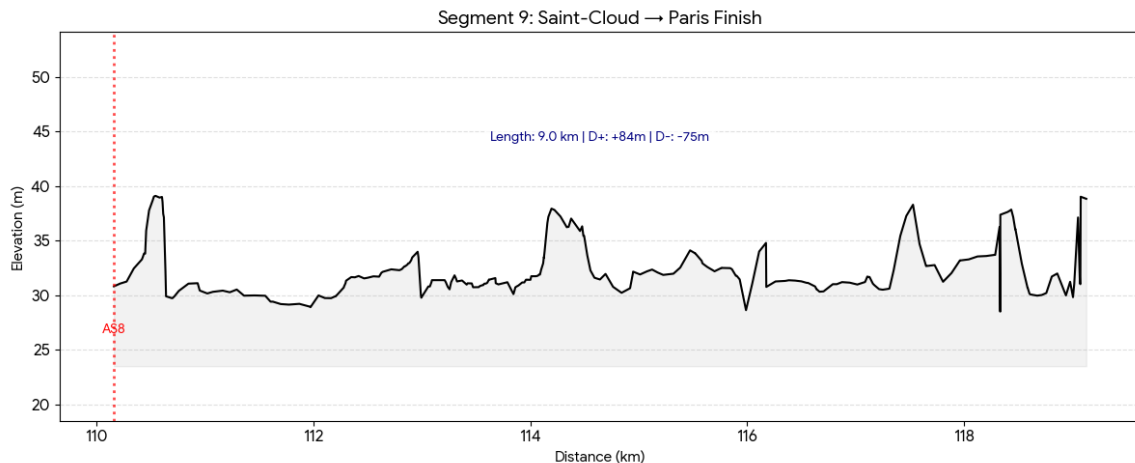
Final Fueling: This is your last chance for significant calorie intake. Ensure you leave AS8 with enough energy to sustain a higher pace on the flat quays.

Mental Clarity: If caffeine, a final dose here can provide the alertness for the finish line navigation.

Shoe Check: Tighten laces if necessary for the final transition to asphalt running.

Up Next: S9: Saint-Cloud → Finish (9.0km). The final flat push through the heart of Paris.

Segment 9: Saint-Cloud → Paris Finish



Dist: 110.2 → 119.1 km

Pace: 5:00–6:00

Time: +12h15 → +13h15

Terrain: Flat Asphalt / Quays

Elev: +84m/-76m

RPE: 5/10

Terrain Analysis

- **Surface Composition:** 80% tarmac and concrete quays, 15% gravel park paths, 5% stairs/ramps.
- **Profile:** Almost entirely flat. The segment consists of a final exit descent from the park followed by a straight-line push along the Seine.
- **Technicality:** Very Low. The primary challenge is the transition back to high-impact asphalt running after 110km of soft soil.

Critical Hazards

- **Asphalt Shock:** The sudden return to hard surfaces can trigger acute joint pain or "locking" of the knees. Maintain a high cadence and avoid long, heavy strides.
- **Mental Relaxation:** The sight of the Eiffel Tower can cause a premature drop in focus. Stay concentrated on the ground to avoid tripping on curbs or urban obstacles.
- **Public Interface:** You are now in central Paris. Be aware of pedestrians, cyclists, and tourists who may not realize you are in the final stages of an ultra-marathon.

Nutrition and Hydration

Nutrition: Minimal intake required. Small sips of water to keep the mouth moist. 1 final gel at KM 115 if needed for a "kick."

Aid Station Actions:

Post-Race Recovery: Start planning your immediate post-finish hydration (Proteins/Carbs) to jumpstart recovery.

Final Push: Use the flat terrain to find a steady, efficient "commuter" rhythm.

Race Complete. *Final target: Eiffel Tower. Celebrate the transition from the deep forest to the iron heart of the city.*

Final Words

You are standing at the threshold of a 120km odyssey.

Ultra-endurance is not merely a test of the heart and lungs, but a masterclass in problem-solving. Over the next several hours, your body will tell you stories of fatigue, doubt, and physical limits. Remember that these are just chapters, not the end of the book.

When the mud of the Chevreuse Valley slows your pace, and the nighttime silence of the Meudon forests tests your resolve, look forward to the quays of the Seine. The Eiffel Tower is not just a finish line; it is a beacon marking the transition from the wild to the heart of Paris.

Trust your training, respect your nutrition protocol, and lean on your crew. They are the silent engine of your success.

Good luck to all the 2026 EcoTrail pioneers. Run with purpose, finish with pride.

About the Author

This roadbook was developed by Benoit Boyadjis, a newcomer in the endurance world training for the 2026 Salomon EcoTrail Paris. Designed as a universal tactical guide, it combines raw GPX data analysis with established physiological protocols to help runners of all levels navigate the 119.1km course efficiently. Good luck !

Appendix A: Pace & Progress Log

Instructions for the Support Crew: Use this matrix to cross-reference the runner's actual elapsed time at each station. Locate the column that matches the current elapsed time to identify the projected finish time and required pace for the upcoming segment.

Strategic Pacing Doctrine

The 120km effort is divided into three metabolic phases. The objective is to manage the transition from road efficiency to trail resistance without over-revving the engine in the first 4 hours.

- **Phase 1: The Efficiency Window (KM 0–40):** Use road running mechanics. Build a time buffer on flat terrain. Target RPE 4/10.
- **Phase 2: Technical Resistance (KM 40–80):** The "Spine" of the race. Constant undulating terrain. Walk the steep climbs to preserve leg muscle integrity for the final 40km.
- **Phase 3: The Urban Crux (KM 80–130):** Hard surfaces return. Focus on posture and core engagement. The goal is to minimize pace decay in the final 26km "Dark Hole."

Tactical Status Indicators

- **ON PLAN:** Arrival within ± 5 minutes of the target column. Maintain protocol.
- **SLIPPING:** Arrival drifts 1 column to the right (>30 mins). Crew must increase caffeine and fast-acting carb intake immediately.
- **CRITICAL:** Arrival drifts 2 columns to the right (>60 mins). Mandatory 5-minute seated reset. Evaluate hydration/salt status.

CREW NOTE: Pacing is the best diagnostic tool for biological health. A sharp drop in segment pace usually indicates a fueling failure, not a lack of fitness.

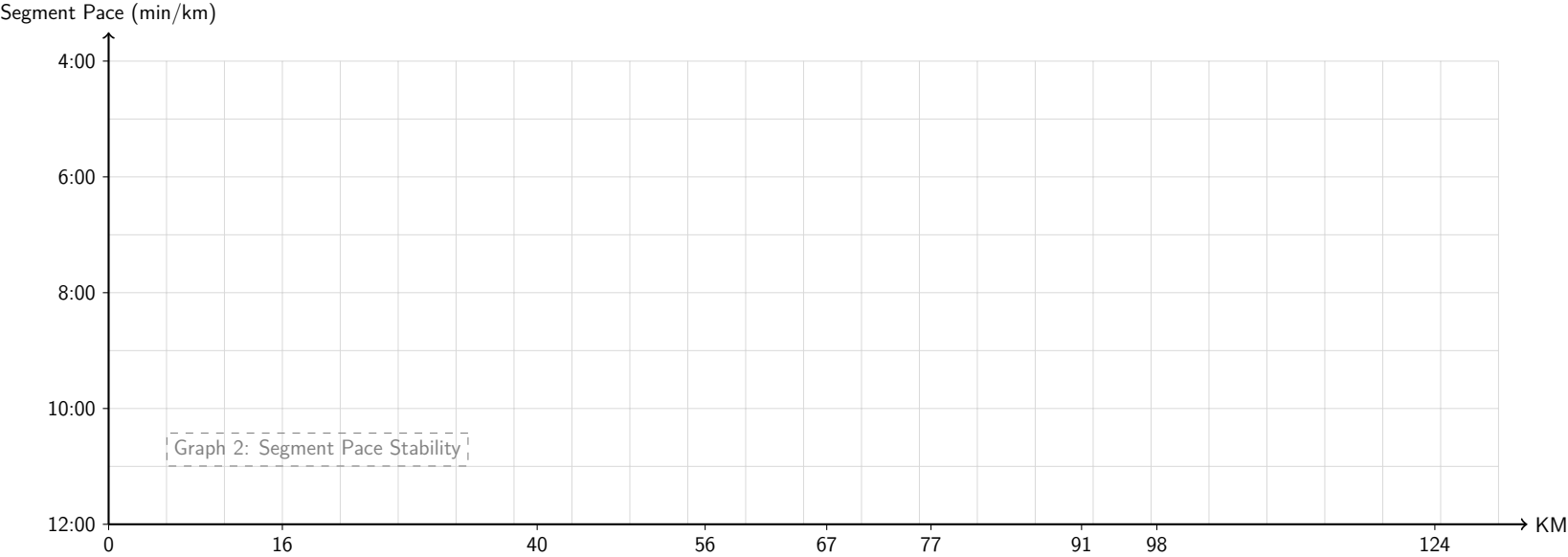
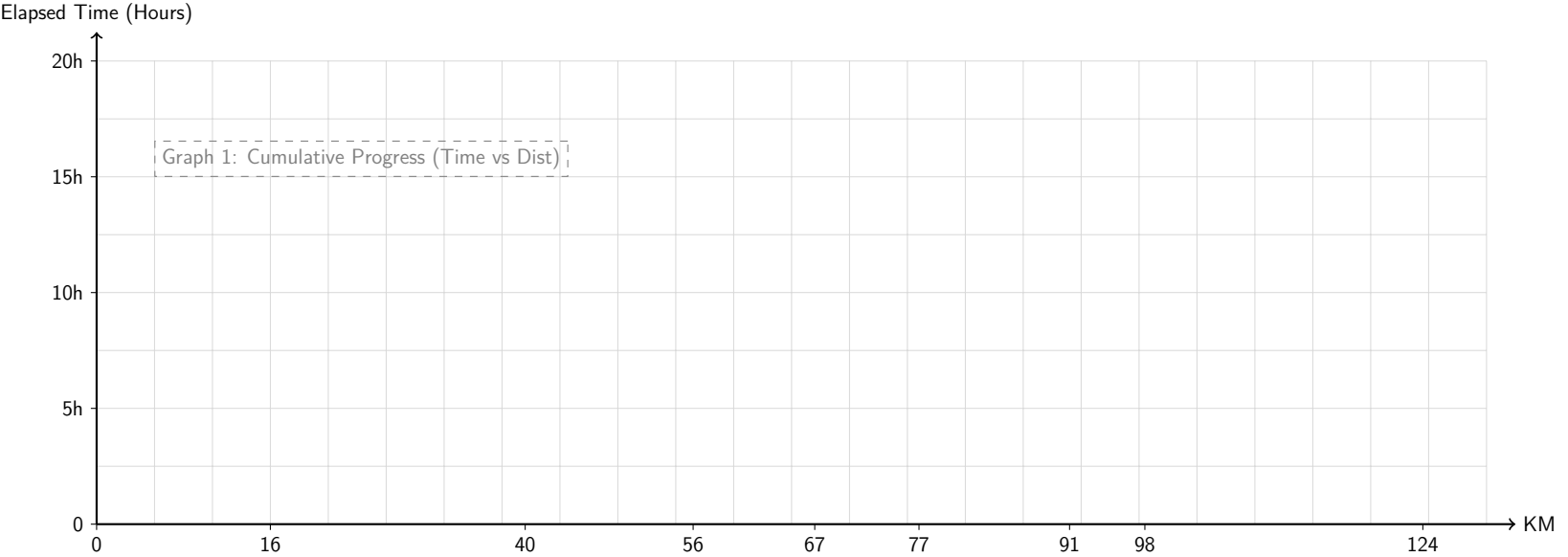
11h – 20h Finish Projections

Format: [Cumulative Elapsed Time] / (Pace for the segment ending at this station)

Station	KM	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h
Start	0	0:00	0:00	0:00	0:00	0:00	0:00	0:00	0:00	0:00	0:00
AS 1	16	1:12 (4:30)	1:18 (4:52)	1:24 (5:15)	1:31 (5:41)	1:38 (6:07)	1:44 (6:30)	1:51 (6:56)	1:58 (7:22)	2:04 (7:45)	2:11 (8:11)
AS 2	40	3:00 (4:30)	3:15 (4:52)	3:32 (5:20)	3:50 (5:47)	4:06 (6:10)	4:22 (6:35)	4:39 (7:02)	4:56 (7:30)	5:12 (7:50)	5:28 (8:20)
AS 3	56	4:26 (5:22)	4:48 (5:48)	5:12 (6:15)	5:36 (6:45)	6:00 (7:15)	6:24 (7:45)	6:48 (8:15)	7:12 (8:45)	7:36 (9:15)	8:00 (9:45)
AS 4	67	5:32 (6:00)	6:00 (6:33)	6:30 (7:05)	7:00 (7:38)	7:31 (8:16)	8:01 (8:55)	8:31 (9:21)	9:02 (10:00)	9:32 (10:33)	10:02 (11:05)
AS 5	77	6:30 (5:48)	7:05 (6:30)	7:40 (7:00)	8:15 (7:30)	8:50 (7:54)	9:25 (8:24)	10:01 (9:00)	10:36 (9:24)	11:11 (9:54)	11:47 (10:30)
AS 6	91	7:45 (5:21)	8:26 (5:47)	9:07 (6:13)	9:50 (6:47)	10:31 (7:12)	11:13 (7:42)	11:55 (8:08)	12:37 (8:38)	13:20 (9:12)	14:02 (9:38)
AS 7	98	8:24 (5:34)	9:09 (6:08)	9:54 (6:43)	10:41 (7:17)	11:27 (8:00)	12:13 (8:34)	12:59 (9:08)	13:46 (9:51)	14:32 (10:17)	15:18 (10:51)
AS 8	124	10:30 (4:50)	11:26 (5:16)	12:22 (5:41)	13:20 (6:09)	14:18 (6:41)	15:16 (7:13)	16:14 (7:46)	17:12 (8:18)	18:11 (8:55)	19:10 (9:30)
FINISH	130	11:00 (5:00)	12:00 (5:40)	13:00 (6:20)	14:00 (6:40)	15:00 (7:00)	16:00 (7:20)	17:00 (7:40)	18:00 (8:00)	19:00 (8:10)	20:00 (8:20)

Note: All values include the 3-minute tactical stop allocation. The (Pace) indicates the speed required to travel from the previous station to the current one.

Performance Visualization Grids



Notes

Appendix B: Cumulative Nutrition & Hydration Log

Instructions for the Support Crew: Use this log to monitor the runner's metabolic health. At every aid station, record the items consumed and update the cumulative totals. Compare the "Actual" intake against the pre-planned "Target." Use the "Diff" column to calculate the surplus (+) or deficit (-). If the energy deficit exceeds 500 kcal, prioritize high-calorie liquids immediately.

Refueling Benchmarks

- **Energy Target:** Aim for an average of **250–300 kcal per hour** to prevent glycogen depletion.
- **Hydration Target:** Aim for **500–700 ml per hour**. Adjust upward if ambient temperature exceeds 18°C.
- **Deficit Rule:** If the energy deficit is large, switch to liquid sugar (Coke/Maltodextrin) as solids may be harder to process late-race.

Electrolytes Tally

Mark an [X] for every salt tablet taken (Recommended: 1 per hour).

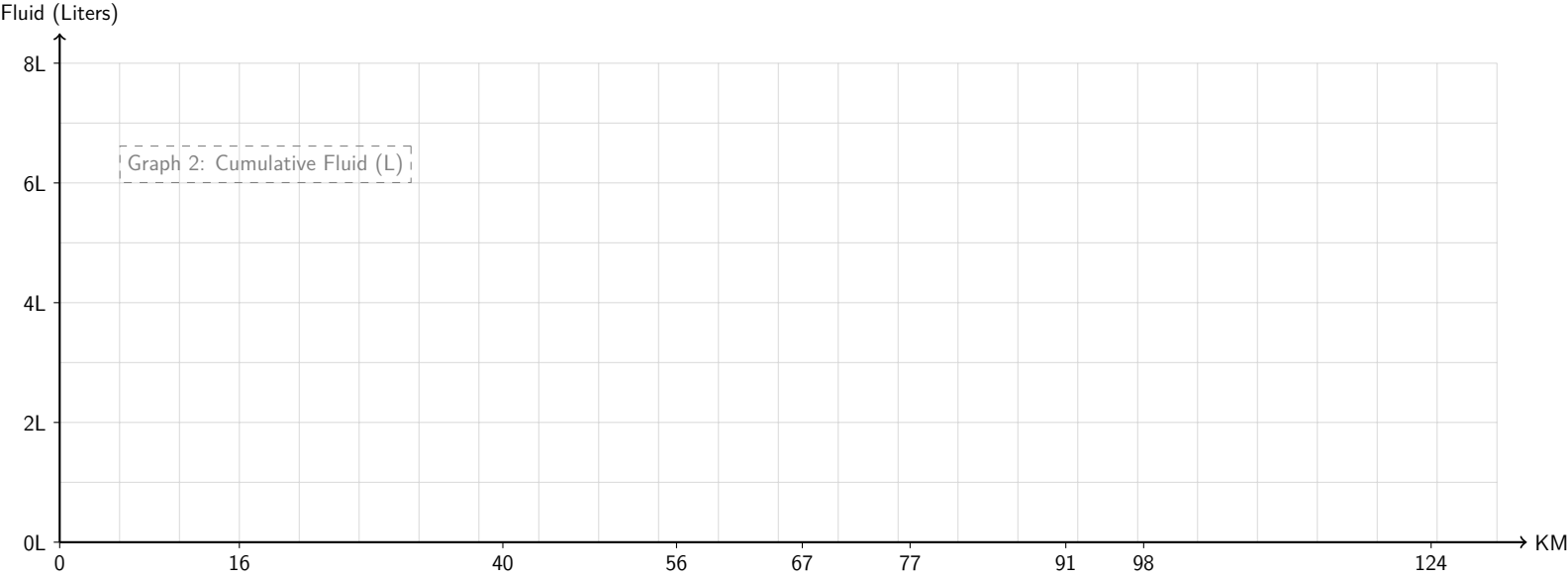
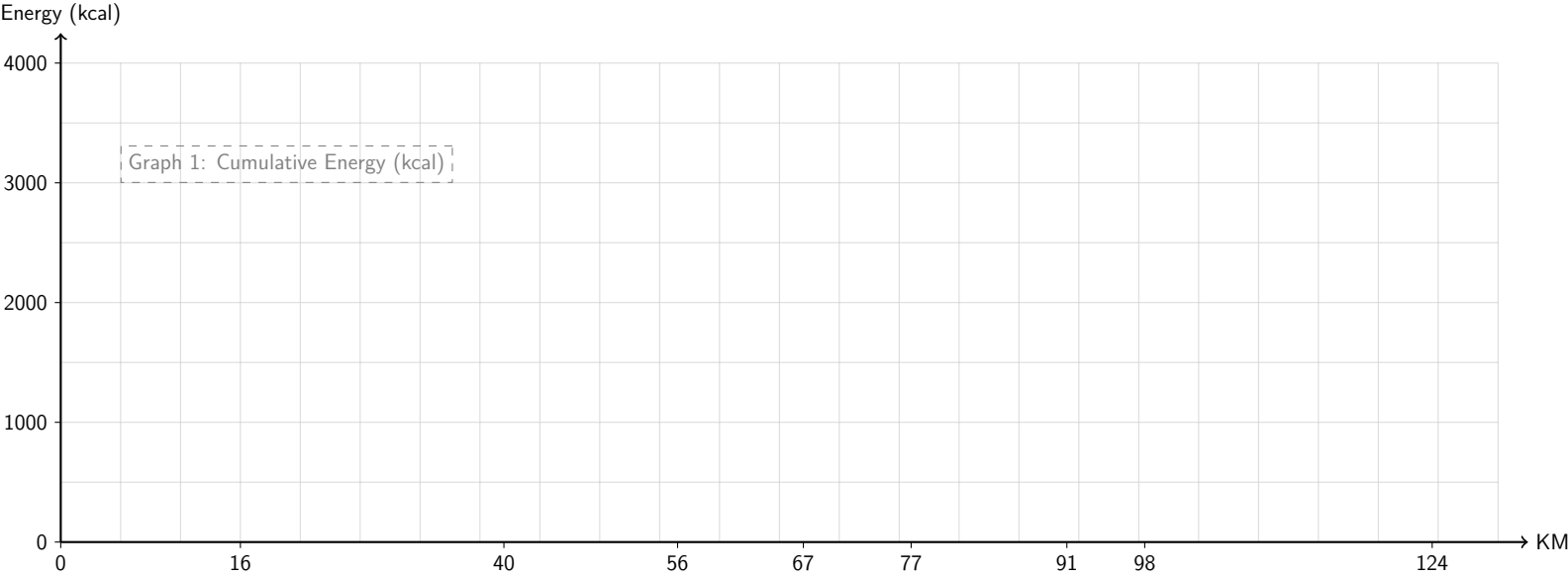
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☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

CRITICAL: A calorie deficit at KM 40 is a biological failure at KM 90. Fuel the machine before the warning light flickers.

Station (KM)	Items Consumed	Cumulative Energy (kcal)			Cumulative Fluid (L)		
		Actual	Target	Diff	Actual	Target	Diff
AS 1 (16)							
AS 2 (40)							
AS 3 (56)							
AS 4 (67)							
AS 5 (77)							
AS 6 (91)							
AS 7 (98)							
AS 8 (124)							

Log Note: Calculate "Diff" as (Actual - Target). Diff's trend shows improvement/degradation.

Metabolic Trend Visualization



Notes

Appendix C: Gear & Equipment Manifest

Instructions for the Support Crew: This document serves as the final logistical safeguard. Equipment management in ultra-endurance is not merely about storage; it is about maintaining the runner's thermal equilibrium and biological integrity through proactive swaps.

1. The Mandatory Equipment Shield

Items designated as **[Mandatory]** are the minimum safety requirements dictated by race regulations. These items must remain in the runner's vest at all times.

- **Verification:** The crew must physically touch or see each mandatory item before the runner exits a support zone.
- **Compliance:** Missing gear results in immediate time penalties or disqualification. There is no room for negotiation on these items.

2. Thermal Transition & Hygiene Strategy

The transition from daylight to darkness is the most dangerous window for a runner's core temperature.

- **The Hot Swap:** Replace damp base layers before the runner feels cold. Once shivering starts, the energy cost to reheat the body is diverted from the legs.
- **Foot Hygiene:** Changing socks at major intervals (e.g., KM 80) reduces friction and skin maceration. Ensure feet are dry and re-lubricated before new socks are applied.

3. Lighting Redundancy Protocol

- **Primary Source:** High-output headlamp for technical forest navigation.
- **Secondary Source:** Emergency backup (lamp or handheld) always present in the vest.
- **Proactive Power:** Swap batteries at the designated support points before they fail. Dimming light causes mental fatigue and increases trip hazards.

CREW PHILOSOPHY: Think like a Formula 1 pit crew. Everything should be uncapped, unwrapped, and ready to deploy 10 minutes before arrival.

Core Equipment Tracking

WARNING - NLDR: this section is too generic (thanks IA); be careful not to align without triple checking on the Ecotrail Website.

Current mandatory gear:

- Réserve d'eau minimum 1,5L Pour rester hydraté tout au long du parcours Pénalité : 2 min
- Réserve alimentaire En-cas type barres ou gels énergétiques
- Gobelet 15cl minimum Pour boire sur les ravitos Pénalité : 30 sec
- Lampe frontale Utile en cas de faible luminosité ou tunnel Pénalité : 2 min
- Brassard réfléchissant Pour rester visible en conditions sombres Pénalité : 2 min
- Couverture de survie En cas d'hypothermie ou blessure Pénalité : 2 min
- Téléphone mobile Avec batterie chargée et numéro à contacter en cas d'urgence Pénalité : 2 min
- Pièce d'identité Format papier ou copie acceptés Pénalité : Disqualification
- Moyen de paiement Pour pallier aux imprévus

Equipment Item	Status	Run.	Crew	Notes
MANDATORY SAFETY KIT (cf. above)				
Survival Blanket (min 1.4x2.1m)	[Mandatory]	☐	☐	
Whistle + Reusable Cup	[Mandatory]	☐	☐	
Waterproof Jacket (Hooded)	[Mandatory]	☐	☐	
Mobile Phone (Roaming On)	[Mandatory]	☐	☐	
Headlamp + Spare Battery	[Mandatory]	☐	☐	
Long-sleeved Base Layer	[Mandatory]	☐	☐	
PRIMARY RACE KIT				
Technical Trail Shoes		☐	☐	
Hydration Vest / Pack		☐	☐	
GPS Watch (Course Loaded)		☐	-	
Spare Socks (Sealed Bags)		-	☐	
NUTRITION & MEDICAL				
Soft Flasks (Total 1.0L+)		☐	☐	
Salt Tablets		☐	☐	
Anti-Chafe Stick / Cream		☐	☐	
Blister Care Kit		-	☐	
		☐	☐	
		☐	☐	

Additional Equipment & Field Notes

Item Description	Qty	Run.	Crew	Notes
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
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		☐	☐	

"Success is where preparation meets opportunity."

Notes

Appendix D: Physiological Troubleshooting

For the Crew: Use this decision tree when the runner is unable to provide clear feedback. Follow the symptoms to the required action protocol.

Crisis A: Gastric Distress & Nausea

Symptoms: Refusing calories, bloating, frequent burping, or vomiting.

ACTION PROTOCOL

1. **The Palate Reset:** STOP all sugar intake for 20 minutes. Give 200ml of *plain* water to rinse the mouth and neutralize the stomach.
2. **Anti-Emetic Support:** If available, provide ginger-based snacks, flat ginger ale, or mint tea to soothe the stomach lining.
3. **Edema Check:** Check the runner's hands/fingers. If puffy (Hyponatremia risk), withhold salt. If not, provide one slow-dissolve salt tablet.
4. **Diluted Fueling:** Upon restart, use only half-strength liquid calorie mixes. Avoid solids until the runner can maintain an easy pace without discomfort.

Crisis B: Hypoglycemia (The "Zombies")

Symptoms: Glazed eyes, slurred speech, unsteadiness, or irrational irritability.

ACTION PROTOCOL

1. **Rapid Glucose:** Administer 150ml of full-sugar cola or a high-glycemic liquid gel immediately. DO NOT use "sugar-free" variants.
2. **The 5-Minute Reset:** Force the runner to sit. Use this time to insulate them with a jacket or space blanket (glucose drops often trigger shivering).
3. **Cognitive Screen:** Perform the "Tracking Test": move a finger in front of their eyes. If they cannot track it or fail to answer "What is the next aid station?", they do not leave the station.
4. **Complex Rebound:** Once the "fog" lifts, provide a small portion of complex carbs (bread/pasta) to prevent a secondary crash.

Crisis C: Acute Muscular Cramping

Symptoms: Sudden sharp pain, localized "locking," or visible muscle fasciculations.

ACTION PROTOCOL

1. **Electrolyte Correction:** Administer a high-concentration electrolyte shot or double the salt tablet dose for the immediate hour.
2. **Topical Intervention:** Apply cooling gel or magnesium spray to the affected muscle. Perform a light, upward "effleurage" massage (no deep tissue work).
3. **Active Recovery:** Do not allow the runner to stay static. Transition to a 10-minute power-walk to allow the muscle to find a new functional equilibrium.

Crisis D: Mental Exhaustion & Dissociation

Symptoms: "I'm done," "I can't see the point," or sustained silence/withdrawal.

ACTION PROTOCOL

1. **The "Next Step" Contract:** Do not discuss the finish line. Say: "Get to the next landmark (5km). If you still want to stop there, we will have a rational conversation."
2. **Sensory Change:** Change one element of their gear (fresh socks, a dry shirt, or a different playlist). Small sensory resets can break a negative feedback loop.
3. **Externalize Achievement:** Remind them of the distance already covered. "You've done 90km; the remaining 30km is just a weekend long run."

Universal Rule: If the runner is moving, they are winning. Only structural failure (sharp bone pain or total inability to bear weight) justifies a DNF.

Appendix E: Crew Navigation Toolbox

This document is for the support vehicle. Your goal is to reach the station, park, and have the 'Pit Stop' ready 15 minutes before the runner's estimated arrival.

Logistics Registry

AS1: Étangs de Hollande

Location: D61, 78610 Les Bréviaires

GPS: [48.7314](#), [1.7978](#)

Transit Challenge: Easy drive from Rambouillet. Forest roads.

Parking: Large dirt lot. 2/10 difficulty.

Crew Task: Runner will be fresh. Focus on checking hydration levels only. No heavy food needed.

AS2: Château de Dampierre

Location: Grande Rue, 78720 Dampierre-en-Yvelines

GPS: [48.7049](#), [1.9874](#)

Transit Challenge: Narrow valley roads. Slow traffic expected.

Parking: Extremely limited. 9/10 difficulty. Arrive early.

Crew Task: Runner will have finished the longest 24km stretch. Check for hot spots/blisters.

AS3: Château de la Madeleine

Location: Chemin de la Madeleine, 78460 Chevreuse

GPS: [48.7101](#), [2.0422](#)

Transit Challenge: Steep access road. Congestion at the top.

Parking: Medium lot. 7/10 difficulty.

Crew Task: High-intensity section. Prioritize electrolytes and mood check.

AS4: Golf National

Location: 2 Rue de la Boulie, 78280 Guyancourt

GPS: [48.7537](#), [2.0777](#)

Transit Challenge: Main roads, high traffic near SQY.

Parking: Optimized parking. 3/10 difficulty.

Crew Task: Halfway point. Possible gear change (dry socks/shirt). Check headlamp batteries.

AS5: Buc

Location: Arcades de Buc, 78530 Buc

GPS: [48.7706](#), [2.1234](#)

Transit Challenge: Residential area. Speed bumps.

Parking: Street parking. 5/10 difficulty.

Crew Task: Nightfall transition. Ensure high-visibility gear is active.

AS6: Jean Jaurès**Location:** Avenue Jean Jaurès, Chaville/Viroflay**GPS:** [48.7975](#), [2.1860](#)**Transit Challenge:** Urban traffic. Watch for race crossings.**Parking:** Urban spots. 6/10 difficulty.**Crew Task:** Fatigue will be high. Use caffeine protocol if agreed.**AS7: Marcel Bec****Location:** Route du Pavillon des Gardes, 92190 Meudon**GPS:** [48.8092](#), [2.2189](#)**Transit Challenge:** Forest entrance. Dark roads.**Parking:** Forest lot. 4/10 difficulty.**Crew Task:** Technical section finished. Check mental clarity. Final 25km push.**AS8: Saint-Cloud****Location:** Grille d'Orléans, 92210 Saint-Cloud**GPS:** [48.8306](#), [2.2226](#)**Transit Challenge:** Tight park entrance. Pedestrians.**Parking:** Public parking. 8/10 difficulty.**Crew Task:** Final resupply. Remove all non-essential gear to lighten the runner for the flat 9km.

Universal Crew Instructions

1. **The '3-Minute' Rule:** Never let the runner stay longer than 3 minutes at a station unless injured.
2. **Trash Management:** Take all wrappers and empty flasks immediately.
3. **The Eyes:** Don't ask 'How do you feel?' - Look at their gait and skin color. Adjust salt/water accordingly.